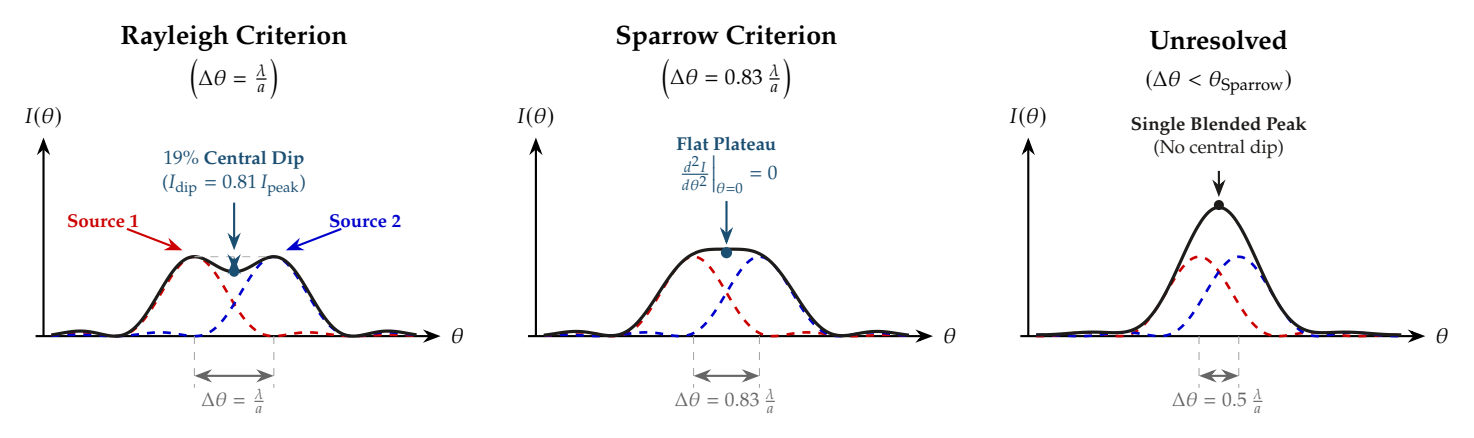

Rayleigh Criterion
$\left(\Delta\theta = \frac{\lambda}{a}\right)$
$I(\theta)$
19% Central Dip
($I_{\text{dip}} = 0.81\, I_{\text{peak}}$)
Source 1
Source 2
θ
$\Delta\theta = \frac{\lambda}{a}$
Sparrow Criterion
$\left(\Delta\theta = 0.83\, \frac{\lambda}{a}\right)$
$I(\theta)$
Flat Plateau
$\left.\frac{d^2 I}{d\theta^2}\right|_{\theta=0} = 0$
θ
$\Delta\theta = 0.83\, \frac{\lambda}{a}$
Unresolved
($\Delta\theta < \theta_{\text{Sparrow}}$)
$I(\theta)$
Single Blended Peak
(No central dip)
θ
$\Delta\theta = 0.5\, \frac{\lambda}{a}$